



CYBERSTER

Internship Program - Batch 2

NetSec Internship - Week 8 Task Report

Network Routing & Switching Concepts

Intern Name:	Muhammad Raza
Instructor Name:	Muhammad Huzaifa Riaz
Company:	CYBERSTER (Batch 2)
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Table of Contents

Section	
1. Theoretical Analysis - Static vs. Dynamic Networking	3
1.1 Static vs. Dynamic VLAN Assignment	3
1.2 Static vs. Dynamic Routing	4
1.3 When to Use Each - Decision Guide	5
1.4 Industry Standard Summary	5
2. Inter-VLAN Routing Lab - Router-on-a-Stick	6
2.1 Lab Topology	6
2.2 Layer 2 Configuration - Switch Setup	6
2.3 Layer 3 Configuration - Router Sub-Interfaces	7
2.4 End Device IP Configuration	8
2.5 Connectivity Verification	9
3. Configuration Summary Checklist	9
4. Conclusion	10

1. Theoretical Analysis - Static vs. Dynamic Networking

In enterprise network engineering, the terms "Static" and "Dynamic" are applied to two completely different configuration paradigms: VLAN Assignment at Layer 2 and Network Routing at Layer 3. Understanding the distinction between these approaches, and knowing when to apply each, is a foundational skill for any network engineer.

1.1 Static vs. Dynamic VLAN Assignment

VLAN Assignment determines how an end-device is placed into a specific broadcast domain (VLAN) at the moment it connects to a switch port.

	Static VLAN (Port-Based)	Dynamic VLAN (MAC / 802.1X-Based)
Concept	The administrator manually assigns a physical switch port to a specific VLAN. Any device plugged into that port is automatically placed in that VLAN regardless of its identity.	The switch port belongs to no VLAN until a device connects. The switch queries a central server (RADIUS/VMPS) that reads the device's MAC address or 802.1X credentials and dynamically assigns the correct VLAN.
Real-World Example	Corporate office: all wall jacks in the accounting department are hardcoded to Switch Ports 1–10, assigned to VLAN 30 (Finance). If an HR printer is plugged in, it instantly joins Finance VLAN regardless of what it is.	Hospital environment: a doctor moves between different wards. Regardless of which ethernet jack they plug into, the authentication server recognises their credentials and dynamically drops the port into the Medical Records VLAN - ensuring access without restricting to a desk.
Administration	Manual configuration per port. Simple to understand and troubleshoot.	Requires RADIUS/VMPS infrastructure. Complex to deploy but fully automated at runtime.
Security	Very secure - port membership is fixed. An unknown device plugged in joins the configured VLAN but gains no special privilege.	Identity-based - extremely granular. A stolen cable plugged into a different port gains nothing without valid credentials.
Scalability	Difficult to scale in large mobile environments - moving devices requires reconfiguring ports.	Scales well in mobile environments - no port reconfiguration needed when users roam.
Use Case	Small-to-medium offices, fixed workstations, printers, servers - environments where users don't move.	Large campuses, hospitals, universities, guest access environments where users roam frequently.
Industry Prevalence	✓ BY FAR the most common. Used in over 95% of enterprise deployments.	Used in large enterprise / healthcare / education environments with roaming users.

★ Industry Standard: Static VLANs are the dominant choice. They are highly secure, trivial to troubleshoot, require zero additional infrastructure, and are the default configuration method taught in every Cisco certification track (CCNA through CCNP).

1.2 Static vs. Dynamic Routing

Routing determines how a router learns the paths (routes) it should use to forward packets between different IP networks, including routing between VLANs and reaching remote sites.

	Static Routing	Dynamic Routing (OSPF / EIGRP / BGP)
Concept	The administrator manually types every route into the router's routing table. The router has no awareness of topology changes - if a link fails, traffic is simply dropped until the administrator manually fixes the route.	Routers use specialised protocols to automatically exchange topology information with each other, dynamically build routing tables, and recalculate the best path if a link goes down - without any human intervention.
Real-World Example	Small retail store with one internal network (VLAN 10) and one ISP link. A single static default route (0.0.0.0/0) pointing to the ISP handles all internet traffic simply and cheaply.	Large university campus with 50 buildings and hundreds of VLANs. When a fibre cable is accidentally cut by construction, OSPF detects the failure within seconds and automatically reroutes student traffic through a backup microwave link.
CPU / Memory	Zero overhead - routes are static entries. Ideal for resource-constrained devices.	Consumes CPU for neighbour hello packets, SPF calculations, and route updates. Negligible on modern hardware.
Convergence	No convergence - if a path fails, traffic is lost until a human intervenes. Recovery time: minutes to hours.	Sub-second to 30-second convergence depending on protocol timers. OSPF BFD can detect failures in <1 second.
Scalability	Only viable for small networks (< 5 routers). Every new network segment requires manual route additions on every router.	Scales to thousands of routers. Adding a new network only requires configuring the local device.
Security	No routing protocol traffic - nothing to attack or spoof. Simpler attack surface.	Routing protocol packets can be targets. Mitigated with MD5/SHA authentication between OSPF neighbours.
Typical Usage	Stub networks (one exit point), home labs, ISP default routes, point-to-point WAN links.	Any enterprise with 3+ routers, multiple sites, redundant paths, or dynamic topology.

★ Industry Standard: Dynamic Routing is the absolute standard in enterprise environments. Static routing is reserved for very small networks, stub sites (one exit point), or specific WAN links where simplicity is preferred.

1.3 When to Use Each - Decision Guide

Scenario	Recommended Choice	Reason
Small office, fixed desks, ≤ 50 users	Static VLANs + Static Routing	Simple, no extra infrastructure needed, easy to maintain
Users roam between floors / buildings	Dynamic VLANs (802.1X) + OSPF	Identity follows the user; routing adapts to topology automatically
Single WAN link to ISP (stub network)	Static Default Route	Only one exit — no routing protocol needed, zero overhead
Multi-site enterprise, redundant WAN links	Dynamic Routing (OSPF/EIGRP)	Automatically fails over to backup link; no human intervention required
University, hospital, large campus (roaming users)	Dynamic VLANs + OSPF	Scale demands automation — static config across 100s of ports is unmanageable
Data center servers, fixed IPs, critical infrastructure	Static VLANs + Static/OSPF	Fixed placement is security best practice for servers; routing choice depends on scale
Home lab or small internship network	Static VLANs + Static/ROAS	Simple to learn, identical concepts, no RADIUS server required

1.4 Industry Standard Summary

Concept	Industry Winner	Why
VLAN Assignment Method	Static	Simpler, universally supported, no additional server infrastructure required. The default in every production environment worldwide.
Routing Protocol	Dynamic (OSPF)	Enterprise networks require automatic convergence, fault tolerance, and scalability. OSPF is the most widely deployed enterprise IGP worldwide.

2. Inter-VLAN Routing Lab - Router-on-a-Stick (ROAS)

To practically demonstrate how traffic moves between isolated broadcast domains (VLANs), a Router-on-a-Stick (ROAS) architecture was deployed in Cisco Packet Tracer. ROAS allows multiple VLANs to be routed through a single physical router interface by dividing it into logical sub-interfaces, each handling one VLAN.

2.1 Lab Topology

Device	Role	Interfaces Used
Cisco 1941 Router	ROAS router - performs inter-VLAN routing via sub-interfaces	Gig0/0 (physical), Gig0/0.10 (VLAN 10), Gig0/0.20 (VLAN 20)
Cisco 2960 Switch	Layer 2 switch - segments PC ports into VLANs and trunks to router	Fa0/1 (trunk), Fa0/2 (VLAN 10 access), Fa0/3 (VLAN 20 access)
PC 1 - HR Department	End device in VLAN 10 (192.168.10.10/24)	Fa0 → Switch Fa0/2
PC 2 - IT Department	End device in VLAN 20 (192.168.20.10/24)	Fa0 → Switch Fa0/3

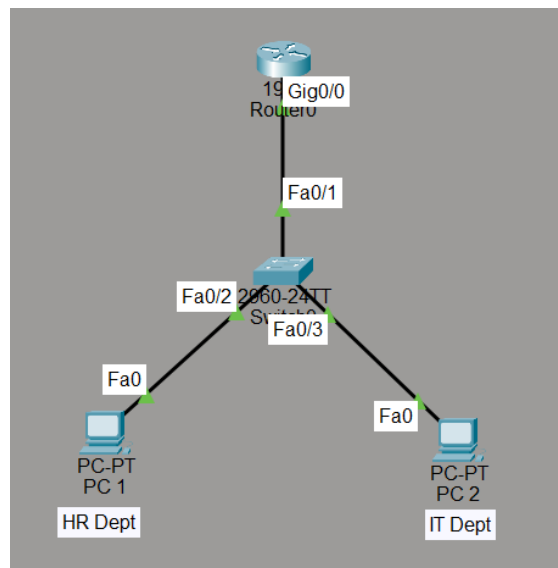
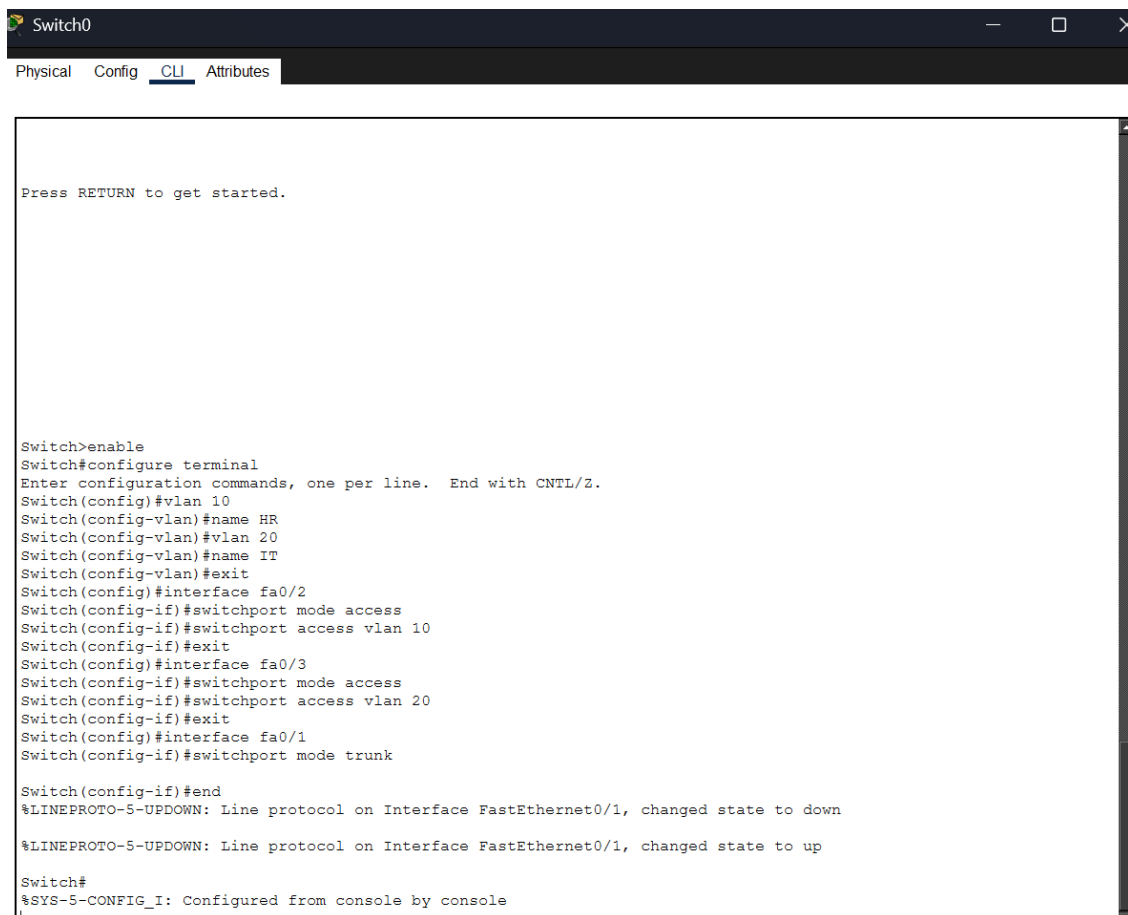


Figure 1: ROAS topology - Cisco 1941 Router connected to 2960 Switch via Gig0/0→Fa0/1 trunk. PC 1 (HR, VLAN 10) on Fa0/2. PC 2 (IT, VLAN 20) on Fa0/3.

2.2 Layer 2 Configuration - Switch Setup

The 2960 Switch requires three configurations: VLAN creation, access port assignment, and trunk port configuration. Without the trunk port on Fa0/1, neither VLAN's tagged traffic can reach the router for routing.

Interface	Mode / VLAN	Purpose
Fa0/2	Access - VLAN 10 (HR)	Dedicated wired port for PC 1 (HR Dept). Only VLAN 10 untagged frames pass here.
Fa0/3	Access - VLAN 20 (IT)	Dedicated wired port for PC 2 (IT Dept). Only VLAN 20 untagged frames pass here.
Fa0/1	Trunk (802.1Q)	Carries both VLAN 10 and VLAN 20 tagged frames on a single cable up to the router. Essential for ROAS.



```
Switch0
Physical Config CLI Attributes

Press RETURN to get started.

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console
|
```

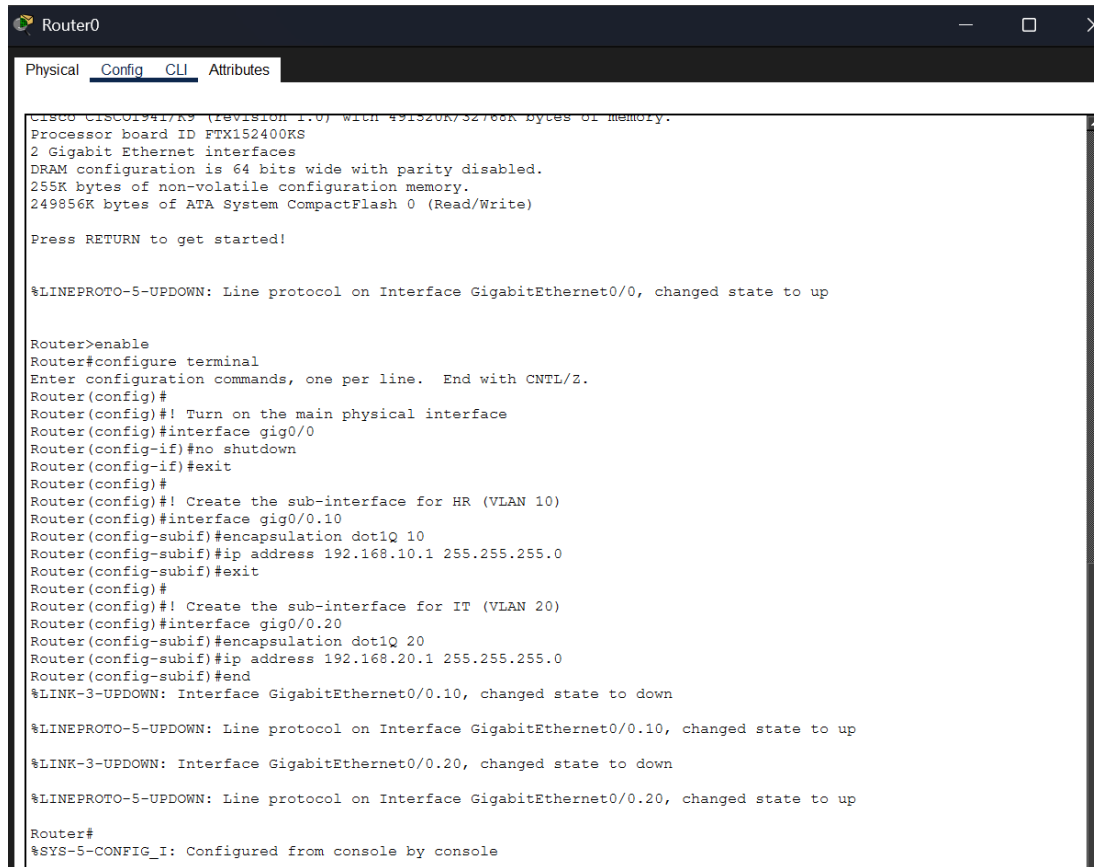
Figure 2: Switch0 CLI - VLAN 10 (HR) created and assigned to Fa0/2; VLAN 20 (IT) created and assigned to Fa0/3; Fa0/1 set to trunk mode. FastEthernet0/1 line protocol confirmed up.

2.3 Layer 3 Configuration - Router Sub-Interfaces

The physical Gig0/0 interface was brought up with no shutdown (no IP address assigned to the physical interface itself - only the sub-interfaces carry IPs). Two logical sub-interfaces were then created:

Sub-Interface	VLAN Tag	802.1Q Encapsulation	IP Address	Serves As
Gig0/0.10	VLAN 10	encapsulation dot1Q 10	192.168.10.1 /24	Default GW - HR Dept
Gig0/0.20	VLAN 20	encapsulation dot1Q 20	192.168.20.1 /24	Default GW - IT Dept

When PC 1 (VLAN 10) sends a packet to PC 2 (VLAN 20), it addresses it to its gateway 192.168.10.1 (Gig0/0.10). The router strips the VLAN 10 tag, routes the packet internally, re-encapsulates it with a VLAN 20 tag, and forwards it out Gig0/0.20. The switch receives the tagged frame and delivers it to PC 2 on Fa0/3. This is the complete ROAS packet flow.



```
Router0
Physical Config CLI Attributes
Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Turn on the main physical interface
Router(config)#interface gig0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Create the sub-interface for HR (VLAN 10)
Router(config)#interface gig0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#! Create the sub-interface for IT (VLAN 20)
Router(config)#interface gig0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#end
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 3: Router0 CLI - Gig0/0 brought up with no shutdown; sub-interface Gig0/0.10 configured with dot1Q 10 encapsulation and IP 192.168.10.1; sub-interface Gig0/0.20 configured with dot1Q 20 and IP 192.168.20.1. Both sub-interfaces confirmed up.

2.4 End Device IP Configuration

Each PC was statically configured with an IP address in its respective departmental subnet. The Default Gateway for each PC is set to its VLAN's router sub-interface IP - this is how the PC knows to forward cross-subnet packets to the router.

Host	VLAN	IPv4 Address	Subnet Mask	Default Gateway
PC 1 - HR Dept	VLAN 10	192.168.10.10	255.255.255.0	192.168.10.1
PC 2 - IT Dept	VLAN 20	192.168.20.10	255.255.255.0	192.168.20.1

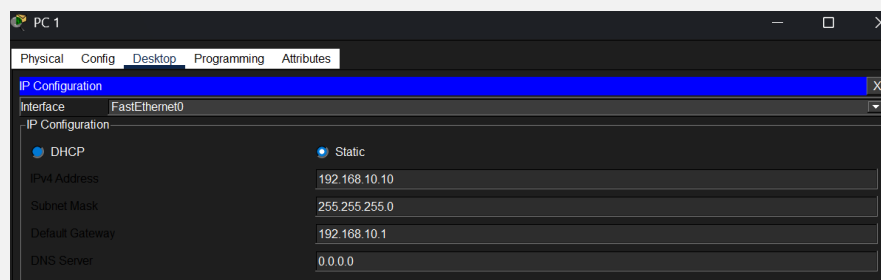


Figure 4a: PC 1 (HR Dept) - Static IP 192.168.10.10/24, Default Gateway 192.168.10.1 (VLAN 10 router sub-interface).

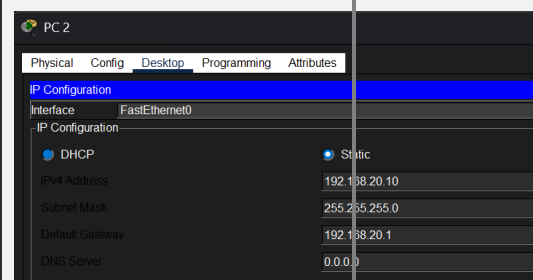


Figure 4b: PC 2 (IT Dept) - Static IP 192.168.20.10/24, Default Gateway 192.168.20.1 (VLAN 20 router sub-interface).

2.5 Connectivity Verification

To verify the ROAS configuration, an ICMP Echo Request (Ping) was initiated from PC 1 (HR Department, VLAN 10) to PC 2 (IT Department, VLAN 20). Without the router, these two VLANs would be completely isolated - this test confirms the router is successfully bridging the broadcast domain boundary.

Traffic Flow Step	Description
Step 1 - ARP	PC 1 sends an ARP broadcast for its gateway 192.168.10.1. The switch tags it VLAN 10 and forwards it up the trunk to the router.
Step 2 - ARP Reply	The router responds with its MAC address. PC 1 now has a Layer 2 path to the gateway.
Step 3 - ICMP to Router	PC 1 sends the ICMP packet. Switch tags it VLAN 10 (via Fa0/2 access port) and forwards it up the Fa0/1 trunk.
Step 4 - ROAS Routing	Router receives the frame on Gig0/0.10, strips the VLAN 10 tag, looks up the routing table, and routes to 192.168.20.0/24.
Step 5 - Delivery to PC 2	Router re-tags the frame with VLAN 20 and sends it down the trunk. Switch delivers it untagged to PC 2 on Fa0/3.

```
PC 1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 5: PC 1 pinging PC 2 (192.168.20.10) - 3/4 packets received, 0ms average RTT, TTL=127. First packet lost to ARP resolution (standard behaviour). Inter-VLAN routing confirmed successful.

Source	Destination	Protocol	Result
PC 1 - 192.168.10.10 (VLAN 10)	PC 2 - 192.168.20.10 (VLAN 20)	ICMP (Ping)	✓ 3/4 - TTL=127, 0ms avg

3. Configuration Summary Checklist

The following checklist confirms every configuration step was completed and verified for the Week 8 lab submission.

✓	Task	Verification
✓	Theory — Static vs. Dynamic VLAN: comparison, real-world examples, and use-case guidance documented.	Completed in Section 1.1 with detailed comparison table
✓	Theory — Static vs. Dynamic Routing: comparison, real-world examples, and industry standard identified.	Completed in Section 1.2 with detailed comparison table
✓	Decision guide for when to use each approach documented with 7 real enterprise scenarios.	Completed in Section 1.3
✓	ROAS topology built in Cisco Packet Tracer: 1941 Router + 2960 Switch + 2 PCs.	Figure 1 - topology confirmed up
✓	Switch configured: VLAN 10 (HR) on Fa0/2, VLAN 20 (IT) on Fa0/3, trunk on Fa0/1.	Figure 2 - Switch CLI confirmed
✓	Router configured: Gig0/0 brought up, sub-interfaces Gig0/0.10 and Gig0/0.20 with dot1Q encapsulation.	Figure 3 - Router CLI confirmed
✓	PC 1 (HR): 192.168.10.10/24, GW 192.168.10.1. PC 2 (IT): 192.168.20.10/24, GW 192.168.20.1.	Figures 4a & 4b - IP config confirmed
✓	Ping from PC 1 (VLAN 10) to PC 2 (VLAN 20) successful - inter-VLAN routing verified.	Figure 5 - 3/4 packets, 0ms, TTL=127

4. Conclusion

Week 8 explored two of the most fundamental design decisions in enterprise networking — how to assign devices to VLANs, and how routers learn paths between networks — and then put those concepts into practice with a live Router-on-a-Stick lab.

Key Takeaways

Static VLANs: The industry default. Port-based assignment is universal, reliable, and requires no additional infrastructure. Every production network uses this as its baseline.

Dynamic VLANs: A powerful identity-based alternative for mobile environments. Requires RADIUS/VMPS infrastructure but enables seamless user roaming across the network.

Static Routing: Best for stub networks and point-to-point WAN links. Simple, zero overhead, but has no fault tolerance - manual intervention required when paths fail.

Dynamic Routing (OSPF): The enterprise standard. Automatically discovers neighbours, shares topology, and converges in seconds after a failure. Non-negotiable for multi-site or redundant networks.

ROAS Lab Result: Successfully demonstrated cross-VLAN communication between HR (VLAN 10) and IT (VLAN 20) using a single router interface divided into sub-interfaces. Ping verified with TTL=127 and 0ms RTT.

The theoretical analysis completed in Part 1 directly explains the practical configuration decisions made in Part 2. Static VLANs were used because the lab represents a fixed-desk office environment. The ROAS technique was chosen over a Layer 3 switch because the Cisco 1941 Router was available and the lab network had only two VLANs - exactly the scenario where a single router interface is the most cost-effective and educationally instructive solution. Together, these concepts form the complete Layer 2-Layer 3 foundation of all enterprise networks covered throughout this CYBERSTER internship programme.